

FIFTH DR. ZAIN MEMORIAL LECTURE
MOLECULAR GENETICS OF HUMAN CANCER:
STATE-OF-THE-ART DIAGNOSTIC AND
THERAPEUTIC APPROACHES

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Cancer is a highly complex genetic disorder characterized by multiple genetic aberrancies that usually accumulate over a long period of time. It is not a single disease, but rather a cumulative name for more than 100 distinct disorders. Despite the fact that each cancer has its own unique features, there are some fundamental abnormalities that are shared by almost all cancers. The quest for an understanding of the basic mechanisms involved in human cancers has provided important insights into the workings of normal cells with highly regulated growth and how these regulatory pathways go awry in cancer cells. An intricate network of signaling pathways normally mediates vital cellular processes such as coordinating proliferation and differentiation, sensing and repairing DNA damage, regulating life span, and promoting cell death or apoptosis. Abnormalities in one or more of these pathways result in uncoordinated and uncontrolled growth, which is a defining feature of cancer cells. During the last several years, there has been a remarkable expansion in our knowledge about the molecular basis of cancer. This review will focus on two central themes in cancer research, i.e. cell proliferation and cell death. A critical balance between cellular growth and elimination by cell death is vital in normal development and in maintaining tissue homeostasis. Defects in either of these two processes contribute to neoplasia (Figure 1). A molecular understanding of cancer will provide new insights for developing methods for early detection and prevention and for designing more relevant therapeutic strategies.

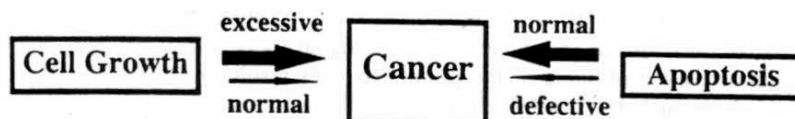


Fig. 1: Cancer, an imbalance between cell growth and apoptosis: Excessive cell proliferation or defective apoptosis contribute to cancer.

A variety of genes have been identified, which constitute important components of the signal transduction circuitry, the alterations of which give a competitive growth advantage to cells. Most of these genes belong to one of the three categories of genes that play a major role in triggering cancer; protooncogenes encode proteins that stimulate cell division, tumor suppressor genes code for proteins that inhibit cell division, and DNA repair genes, the protein products of which are involved in repairing various types of DNA damage.

*Dedicated to the fond memory of my teacher and mentor, Late Dr. M. Zain-ul-Abedin (1932-1990).

Protooncogenes:

The genes implicated in the development of cancer were first encountered in retroviruses. These retroviral oncogenes apparently arose from cellular genes, termed protooncogenes, during the life cycle of the virus in the infected host. The proteins encoded by protooncogenes appear to be important for cellular growth and differentiation and are expressed in a highly regulated fashion at specific times and in selective tissues. There is an extremely long list of protooncogenes, which are very diverse in nature and, when activated, participate in cellular transformation by driving excessive cell proliferation forward. The mechanisms of activation of protooncogenes often encountered in human tumors include point mutation, amplification, enhanced transcription without amplification, and chromosomal translocation (Figure 2). The net result of these alterations is either the expression of an altered gene product or increased expression of the normal gene product with deleterious consequences for the cell.

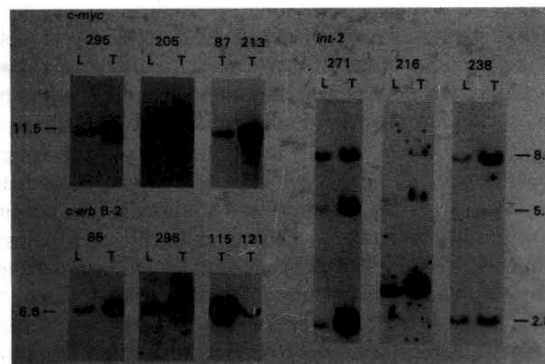


Fig. 2: Amplification of protooncogenes (c-myc, int-2; and c-erbB-2) in human breast tumors: L, lymphocyte DNA; T, tumor DNA. Numbers on the top indicate patient numbers and numbers on the sides indicate the sizes of the fragments in kilobases.

Protooncogenes exhibit a dominant phenotype at the cellular level; the altered copy represents a gain of function mutation, overrides the function of its normal allele, and is sufficient to produce an oncogenic effect. Generally, protooncogenes code for cellular proteins that are involved in a wide variety of signal transduction pathways (Table 1). Mutational activation of some of the protooncogenes results in overproduction of their protein products such as growth factors or growth factor receptors with tyrosine kinase activity involved in autocrine or paracrine signaling pathways. Point mutations in some protooncogenes convert their protein products into oncogenic proteins; examples include members of the ras gene family that are guanine nucleotide-binding proteins and participate in stimulatory signal transduction pathways. Overproduction of yet other protooncogenes like members of myc family, which modulate transcription, also contributes to unrestrained growth. Still other proteins that play important roles in cell cycle regulation and blocking cell death pathways are often overexpressed in various malignancies. In summary, it appears that protooncogenes encode for proteins that function in various locations within the cell, i.e. cell surface membrane, cytoplasm, and nucleus. Furthermore, crucial functions are ascribed to these genes such as DNA replication, transcriptional regulation, guanine nucleotide-mediated signal transduction, and protein phosphorylation. Perturbation of any of these functions is therefore, understandably, among the underlying causes of oncogenesis.

Table 1: PROTOONCOGENES**Growth Factors/Growth Factor Receptors***EGFR, PDGFR, erbB, erbB2, & RET***Guanine Nucleotide-Binding Proteins***H-ras, K-ras, & N-ras***DNA-binding Proteins***c-myc, N-myc, & L-myc***Misc***Bcl-2, cyclin D1, & MDM2***Tumor Suppressor Genes:**

As the name indicates, tumor suppressor genes function to suppress neoplastic growth of cells. Mutations in these genes impair this growth-suppressor mechanism by removing the brakes from the cell cycle resulting in a higher rate of uncontrolled growth. Tumor suppressor genes were first identified in inherited cancer syndromes, where predisposition to cancer is often the result of inheritance of recessive mutant alleles at a specific genetic locus. The inactivation of tumor suppressor genes in human tumors was proposed by Alfred Knudson in 1971 when he described his landmark "two hit" model where both "hits" or mutations target the same gene. According to this concept, predisposed individuals inherit one mutant copy of the gene in their germline. The second mutational event occurs somatically, often in the form of loss of the chromosomal region carrying the target gene, resulting in a reduction to homozygosity of the target gene as well as the flanking genes. Sporadic or nonheritable tumors develop in individuals, who do not carry germline mutations and therefore require two independent somatic mutations in both copies of the relevant gene (Figure 3). Thus, this theory unites both the heritable and nonheritable or sporadic forms of cancers.

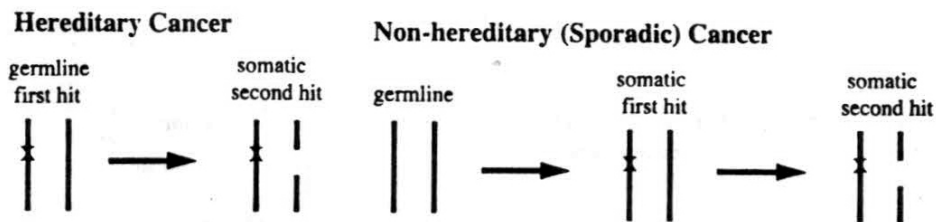


Fig. 3: Two-Hit model of inactivation of tumor suppressor genes: In hereditary cancers, the first hit is present in the form of a germline mutation in one copy of the tumor suppressor gene. In predisposed individuals, second copy of the tumor suppressor gene is inactivated by somatic mutation. In sporadic cancers, both copies of the tumor suppressor gene are inactivated by two independent somatic mutations.

Knudson's hypothesis, which was originally proposed to explain the natural history of retinoblastoma, the embryonal tumor of the developing retina, also predicts that individuals with germline mutations of tumor suppressor genes are at a far greater risk of developing tumors than the general population. This is because the probability of acquiring a single somatic mutation, when the first mutation is already present in the germline, is exponentially greater than the probability of acquiring two independent somatic mutations in the same gene. It also became quickly evident that homozygous functional inactivation of the gene is involved in both inherited and sporadic forms of retinoblastoma. Furthermore, in the hereditary form of the disease, where children inherit one mutant copy of the gene, multiple bilateral tumors develop at an early age. In the nonhereditary form, on the other hand, because of the rarity of the occurrence of two mutations targeting the same gene in a single cell, a unilateral single tumor develops at a later age.

There is persuasive evidence for the functional inactivation of tumor suppressor genes in many human neoplasms. The cloning of the retinoblastoma gene, Rb, provided a conceptual basis to search for tumor suppressor genes in both familial and sporadic cancers carrying frequent deletions on specific chromosomes. This classical paradigm of tumor suppressor genes, first proven in retinoblastoma, has now been extended to many other human cancers and has become a central dogma in cancer genetics. Loss of specific chromosomal regions, which is frequently observed in human malignancies, indicates the presence of tumor suppressor genes in the deleted areas of the chromosomes and is helpful in isolating the gene sometimes by the positional cloning approach. More than 20 tumor suppressor genes have been cloned so far from a variety of tumors by using diverse strategies.

In stark contrast to the dominant nature of mutations in protooncogenes, tumor suppressor genes are recessive in the heterozygous state and their presence is masked by the normal allele. Also, as opposed to activating and gain of function mutations of protooncogenes, mutations inactivate tumor suppressor genes and result in the loss of function (Table 2). However, in a manner similar to protooncogenes, the protein products of tumor suppressor genes function in all locations in the cell, i.e. at the cell surface, in the cytoplasm, and in the nucleus and are involved in a variety of signal transduction pathways. Examples of two tumor suppressor genes, p53 and PTEN, elucidate their pivotal roles in various molecular pathways, which, when deranged by mutations, lead to the development of a wide variety of malignancies.

Table 2: Characteristics of protooncogenes and tumor suppressor genes.

Protooncogenes	Tumor Suppressor Genes
• promote cell growth	• inhibit cell growth
• active when mutated	• inactive when mutated
• dominant in nature	• recessive in nature

p53: Inherited mutations in the p53 gene are believed to be the underlying cause in many cases of Li-Fraumeni syndrome, which is a rare autosomal dominant disorder characterized by malignant transformation in a wide spectrum of target organs among which breast carcinomas are most frequent. The p53 tumor suppressor gene is located on chromosome 17p13.1 and encodes a multifunctional nuclear phosphoprotein composed of 11 exons and 393 amino acids. There are five regions of the gene that are conserved across the species. The protein can be divided into several structural and functional domains including the N-terminal transactivation domain, the central core domain with sequence-specific DNA binding activity, and the C-terminal DNA-damage recognition domain. In about 50% of all human cancers, the p53 gene is homozygously inactivated by missense mutations coupled to the loss of the wild type allele (Figure 4). There are about 9000 p53 mutations reported in the database so far. The prevalence of point mutations along with their scattered distribution appears to be a unique feature of p53 as many other tumor suppressor genes, including Rb, APC, BRCA1, and PTEN, are inactivated by frameshift or nonsense mutations truncating the proteins. A vast majority of point mutations in the p53 gene occur in the central core domain between exons 5 and 9 resulting in the loss of DNA binding.

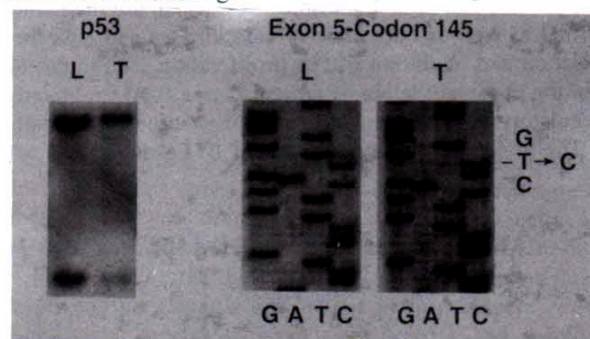


Fig. 4: Homozygous inactivation of p53 gene in human glioblastoma: L, lymphocyte DNA; T, tumor DNA. Absence of the lower band in tumor DNA indicates loss of heterozygosity or loss of one copy of the p53 gene and the sequence of the remaining copy shows a mutation. Note transition of T to C in codon 145 of exon 5 in the tumor DNA compared to the normal sequence in the lymphocyte DNA.

The protein appears to perform diverse cellular functions. The sequence-specific DNA binding and transactivation are key biochemical activities that mediate the biological effects of the p53 protein. Among its functions are transcriptional transactivation as well as transcriptional repression and a role in DNA replication. However, its more global and crucial roles include 1) G1/S checkpoint control regulating cell division, 2) universal sensing of genotoxic stress culminating in DNA damage response to maintain genomic integrity, 3) induction of cell death or apoptosis, and 4) a negative regulatory role on angiogenesis by upregulating angiogenesis inhibitors like thrombospondin and downregulating angiogenic inducers such as vascular endothelial growth factor (VEGF). Introduction of wild type p53 has been shown to inhibit growth of tumor cells in culture and can prevent tumor formation in animal models. p53 has such significant effects on growth regulation and cell signaling that it is perhaps not surprising that it has been implicated in a wide spectrum of malignancies. It is in fact the single most frequently mutated gene in human cancers and therefore lends itself as an excellent target with a very high therapeutic index.

PTEN: PTEN (also known as MMAC1 for mutated in multiple advanced cancers) is the first tumor suppressor with a defined enzymatic activity. The gene is located on chromosome 10q23.3 and encodes a predicted protein product of 403 amino acids containing a protein tyrosine phosphatase domain found in dual specificity phosphatases. The name PTEN reflects the presence of the protein tyrosine phosphatase domain as well as the domain with homology to a cytoskeletal protein, tensin. The gene is mutated in the germline of patients with rare autosomal dominant cancer syndromes, Cowden's disease and Bannayan-Zonana syndrome predisposing to multiple cancers. Somatic mutations in the PTEN gene are frequently found in a variety of sporadic human tumors in which the wild type allele of the gene is inactivated mostly by deletions thus conforming to the classical paradigm of tumor suppressor genes.

PTEN dephosphorylates the second messenger phosphatidylinositol 3,4,5-triphosphate (PIP3), an activity that may be critical for growth regulation. Phosphorylation and dephosphorylation are key regulatory mechanisms in a variety of fundamental cellular processes including cellular proliferation and differentiation. A critical balance among these processes is maintained by protein kinases, proteins widely implicated in carcinogenesis, and protein phosphatases, which are the natural antagonists of the action of normal as well as oncogenic protein kinases and have been associated with cell cycle regulation and inhibition of cell proliferation. The intracellular levels of PIP3 directly control the phosphorylation of protein kinase B/Akt protooncogene, which is one of the key regulatory molecules of cell survival in response to multiple apoptotic stimuli (Figure 5). The growth-inhibitory function of PTEN is evident when wild-type

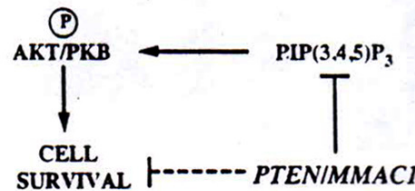


Fig. 5: PTEN as a negative regular of cell survival: Protein kinase B (PKB) or AKT is implicated in protection from apoptosis and cell survival and its activity is directly dependent on intracellular PIP3 levels. PTEN inhibits AKT/PKB activity by dephosphorylating PIP3 and thus negatively regulates cell survival.

cDNA is transfected into a variety of cancer cell lines including glioblastoma, melanoma, and renal, breast, and prostate carcinomas. Mutant forms of PTEN that retain protein phosphatase but lose lipid phosphatase activity fail to suppress the growth of glioma cell lines. In addition to modulating signal transduction and cell cycle progression, phosphorylation affects other cellular processes such as cell-cell and cell-matrix adhesion, migration, and angiogenesis, processes that are important not only for normal cellular physiology, but also impact tumor growth, invasion, and metastasis. Inactivation of the PTEN gene in human cancers may therefore contribute not only to proliferation but also to invasive potential of tumors.

DNA Repair Genes:

During the replication phase of the cell cycle, three billion bases of the haploid human genome are duplicated. To maintain the fidelity of this replication process, several

genetically controlled and elaborate mechanisms have evolved that function as “spellcheckers” to repair the damage to DNA. In the case of DNA damage, which may be a consequence of endogenous random errors of the DNA replication machinery and/or exogenous insults caused by radiation, chemicals, or ultraviolet light, specific mechanisms for repairing specific errors are activated. Conceivably, mutations in the genes encoding the components of this DNA repair machinery involved in proofreading and correcting random and/or specific types of errors result in a replication error positive or “mutator” phenotype. One manifestation of this phenotype is errors in accurate replication of the repetitive di-, tri-, and tetranucleotide repeats, which are scattered throughout the genome, leading to genetic instability.

Defects in four prototype DNA mismatch repair genes, hMLH1, hMSH2, hPMS1, and hPMS2, were identified in the germline of patients from hereditary nonpolyposis colorectal cancer (HNPCC) families. HNPCC family members develop colonic tumors as well as tumors of various other organ sites characterized by a high level of genomic instability. Other disorders associated with faulty DNA repair genes in the germline include Bloom’s syndrome, Fanconi’s anemia, ataxia telangiectasia, and xeroderma pigmentosum.

Gatekeeper, Caretaker, and Landscaper Genes:

The identification of cancer-susceptibility genes has taught us a great deal about the molecular mechanisms of carcinogenesis. The concept of gatekeeper, caretaker, and landscaper genes characterized by their contribution to the neoplastic process was recently introduced by Bert Vogelstein and his colleagues. There are a variety of tumor suppressor genes, which are almost always inactivated in early precursor lesions suggesting that their inactivation may be an essential and rate-limiting step. Thus, inactivation of these genes contributes directly to the initiation and development of specific neoplasms and they appear to function as “gatekeepers” to prevent neoplastic growth. Each cell type appears to have a gatekeeper and inactivation of a given gatekeeper leads to a very specific spectrum in terms of tissue distribution of cancer. It is therefore not surprising that introduction of the normal copy of the missing gatekeeper gene into tumor cells suppresses their neoplastic growth. Examples of gatekeeper genes include Rb in retinal epithelial cells, APC in colon epithelial cells, VHL in kidney cells, NF1 in Schwann cells, and PTC in skin epithelial cells.

Genes involved in maintaining genetic stability appear to suppress neoplasia indirectly and hence their inactivation is not required for neoplastic growth. The DNA repair genes belong to this class of indirect tumor suppressors, the protein products of which function as “caretakers” of the genome. Breast cancer susceptibility genes, BRCA1 and BRCA2, which are probably involved in monitoring and repairing DNA double strand breaks, are also considered as caretaker genes. Although inactivation of a caretaker gene results in a greatly increased mutation rate genomewide, restoration of its function to a cancer cell does not affect its neoplastic growth. Some tumors develop because of abnormal cellular microenvironment or “landscaping” defects. For example, inherited mutations in the PTEN or SMAD4 genes lead to the development of hamartomatous growths providing an abnormal environment for tumor growth. However, it is possible that landscaper genes, in other situations and cell types, function as gatekeepers.

Cell Cycle:

A defining feature of cancer cells is to ignore growth inhibitory signals and continue dividing under conditions that do not permit the growth of normal cells. During normal cell division, a molecular network of enormously intricate and regulated events and feedback mechanisms, generally referred to as the cell cycle clock, controls the replicative process. The cell cycle clock is comprised of four major phases termed as G1 (first gap phase in which cell increases in size and prepares to duplicate its DNA), S (synthesis of DNA with precise duplication of chromosomes), G2 (second gap phase in which cell prepares for the final phase), and M (in which the parent cell divides to produce daughter cells) (Figure 6). Built into this complex network are transition check

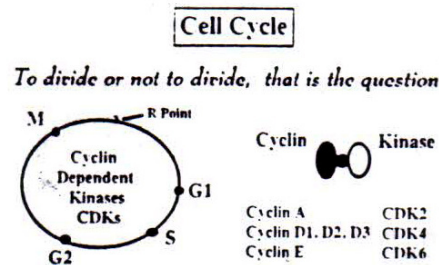


Fig. 6: The cell cycle clock: The G1, S, G2 and M phases and the restriction point R in the early G1 phase are indicated. Members of the cyclin and kinase families assemble in different combinations at each stage of the cell cycle and are intricately regulated by a myriad of extra and intracellular factors.

points between the phases, where surveillance mechanisms ensure the successful completion of earlier events and ascertain the integrity of the genome before permitting the cell to move on to the next events. The cell is particularly receptive to the extracellular growth regulatory signals in the early G1 phase of the cell cycle. If the integrated response of growth stimulatory and growth inhibitory signals is positive, a commitment is made to proceed through the cell cycle and the cell passes an important checkpoint known as the restriction or the R point, which is in effect a point of no return. After the R point transition, the cell is relatively oblivious to the negative growth regulatory signals.

The various checkpoints of the cell cycle are regulated by cyclin dependent protein kinases (CDKs), a family of heterodimeric proteins consisting of the catalytic kinase and the regulatory cyclin subunits (Figure 6). Unique combinations of CDKs and cyclins assemble during each phase of the cell cycle and drive the process of cell proliferation forward by phosphorylating specific key targets. The activity of CDK/Cyclin complex is constrained by binding to a diverse family of inhibitors. The unconstrained proliferation can largely be explained by the gain of positive regulators or the loss of negative regulators of the cell cycle machinery. Cancer, in essence, appears to be a disease of the cell cycle. As a group, mutations in a variety of cell cycle genes probably constitute the most common genetic abnormality in cancers. Understandably, both protooncogenes and tumor suppressor genes are centrally involved in the control of cell division. For example, the Rb protein, in its hypophosphorylated form, exerts a growth suppressive effect by complexing with the transcription factor E2F. In a subset of breast cancers,

cyclin D1 is amplified and overexpressed thus assuming the role of an oncogene and initiating a chain of molecular events that facilitate G1-S transition. Cyclin D1 binds to and activates CDK4, which then phosphorylates the Rb protein releasing E2F from its inhibitory constraints thereby enabling it to activate the transcription of genes required for DNA replication. Similarly, two other tumor suppressors, p16 and p15, display specificity primarily for CDK4 and inhibit transition at the G1 phase of the cell cycle (Figure 7).

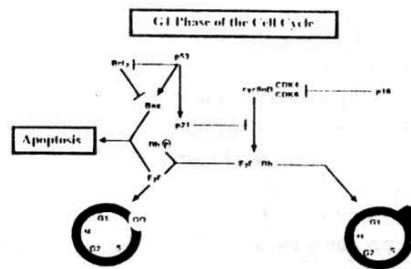


Fig. 7: Regulation of the G1 phase of cell cycle by multiple oncogenes and tumor suppressor genes: Cyclin D/CDK4 or CDK6 complex, which is under the negative regulation of p16/p15 and p21, phosphorylates Rb thereby releasing the transcription factor E2F to transcribe the genes needed for the progression of the cell cycle. p53 arrests cell cycle progression by stimulating the transcription of p21, an inhibitor of cyclin/CDK complexes at G1 and other checkpoints. p53 also sensitizes cells to apoptosis by stimulating proapoptotic Bax and downregulating antiapoptotic Bcl2. Bax and 2F also cooperate to induce apoptosis.

The tumor suppressor p53 plays a central role in the cell cycle and is involved at multiple check points. Its downstream effector, p21, is a universal inhibitor of many CDKs (figure 7). The p53 protein also appears to be a major component of the DNA damage response machinery. Multiple signaling pathways monitoring the genetic damage converge on p53, which, by some unknown mechanisms, senses the damage. The cell cycle is then arrested to allow the DNA-repair machinery enough time to correct the damage. However, if the damage to the DNA is extensive and cannot be repaired, then p53 triggers cell death (Figure 8). Thus, there are two processes, cell growth and cell death, which are influenced by p53 and which when defective, directly contribute to cancer.

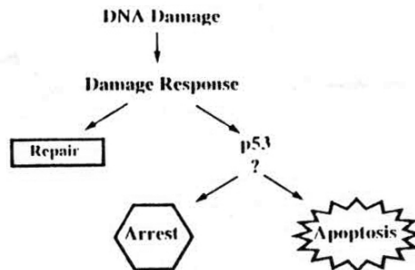


Fig. 8: Role of p53 in DNA damage response: Damage to the DNA is sensed by p53-dependent and non-p53-dependent machinery and repaired. p53 mediates both growth arrest and apoptotic response pathways.

Apoptosis:

Programmed cell death, also known as apoptosis, is an evolutionarily conserved process and along with inhibition of cell proliferation, plays a critical role in maintaining normal tissue homeostasis during development and differentiation. Cells are routinely eliminated during morphogenesis in the developing embryo and during the physiological turnover in adult tissues. Apoptosis is frequently dysfunctional in cancerous cells as a consequence of which cells with extensive and irreparable DNA damage cannot be eliminated.

Apoptosis is carried out through the activation and function of a family of highly conserved cysteine-containing aspartate-specific proteases known as caspases. Although caspases are constitutively expressed in most cells, they are present in the cell as proenzymes and their activity is intricately and effectively regulated by a multitude of proapoptotic and antiapoptotic factors (Figure 9). Thus, like many other complex physiological phenomena, very often the balance between competing pro- and antiapoptotic activities determines cell fate. The activation of caspases proceeds via three different highly complex mechanisms that can be, for the sake of brevity, described as follows: 1) processing by other proteases including caspases: procaspase aggregation is an important step which initiates the cleavage thus converting the proenzyme into its active form, 2) activation via interaction with death receptors and their associated molecules: some apoptosis signaling pathways lead to expression of death receptor ligands, which then engage their receptors and recruit procaspases, and 3) cytochrome c-dependent activation: various forms of cellular stress trigger mitochondrial release of cytochrome c, which then combines with Apaf-1 (apoptotic protease activating factor) and initiates caspase activation.

Apoptosis is negatively regulated by members of the Bcl-2 family and IAP (inhibitor of apoptosis) which directly inhibit the terminal caspases (Figure 9). Recently, an IAP-related protein, survivin, was found to be expressed at high levels in transformed cell lines and most human cancers including those of lung, colon, breast, prostate, pancreas, skin, and brain. Transient transfection of antisense survivin in a cervical carcinoma cell line resulted in induction of apoptosis. These findings suggest that survivin can serve as a potential therapeutic target in cancers and can be exploited to induce death of cancer cells.

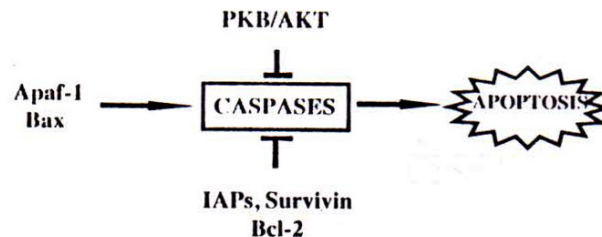


Fig. 9: Caspases and cell death: Caspases are activated by various mechanisms some of which involve the participation of proapoptotic factors such as Apaf-1 and Bax2. PKB/AKT, a strong cell survival regulator and many other antiapoptotic factors including IAPs, Bcl2 and survivin inhibit caspases and thus apoptosis.

Senescence:

Quite distinct from the apoptotic program, normal cells have another built-in defense against uncontrolled proliferation. Human cells, when grown in culture, have a finite life span and senesce following a specific number of cell doublings. It has now become clear that telomeres serve as a molecular device that keeps track of the number of cell doublings. Telomeres are the extreme ends of the eukaryotic chromosomes and are composed of tandemly repeated specialized sequences implicated in maintaining chromosome stability and providing protection against illegitimate recombination and enzymatic degradation. The telomere repeat unit in humans is the hexanucleotide TTAGGG and is about 5-12 kb long in somatic cells. All chromosomes lose a small amount of telomeric DNA during each cell division, which is a natural consequence of the way the cellular DNA machinery works (Figure 10). This so called "end replication problem" ensures progressive shortening of telomeres, which is believed to function as a "mitotic clock" eventually signaling senescence when a critical threshold level is reached.

END REPLICATION PROBLEM

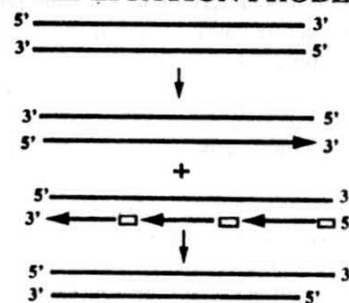


Fig. 10: The End Replication Problem (ERP): The DNA polymerase replicates the leading strand continually. The lagging strand, however, is replicated in short segments called Okazaki fragments, which are initiated from several RNA primers (small boxes). These primers are finally degraded and the gaps filled in by using the upstream Okazaki fragment as a template. Removal of the terminal RNA fragment leaves a gap because of the absence of the template beyond the 5' end thus creating the 'end replication problem.'

Telomerase is the enzyme that cures the end replication problem. It is a ribonucleoprotein with a unique reverse transcriptase that carries its own RNA moiety and uses it as a template to fill in the gaps at the telomeric ends created by the DNA polymerase. Most adult human cells are lacking in telomerase activity and therefore have a finite life. In contrast, telomerase activity is detected in immortalized cell lines and in about 90% of human cancers. Immortalized cells in culture escape senescence and acquire the capability of proliferating indefinitely probably by reactivating or upregulating telomerase and thus stabilizing the chromosomal ends. The upregulation of the telomerase activity is thought to confer upon cancer cells the proliferative capacity to accumulate necessary mutations to become malignant.

The RNA component of human telomerase (hTR) and more recently its catalytic subunit (hEST2/hTERT) have been cloned thus providing targets for blocking the

enzymatic activity. As the molecular understanding of the basic biology of telomerase, especially of its regulation, grows, it will be possible to design highly targeted and effective antitelomerase approaches to limit the replicative capacity of the cells, which in itself is considered a tumor suppressor mechanism.

Translation of Research into Diagnostic and Therapeutic Applications:

Much progress has been made in understanding the genetic basis of various human cancers. The rapidly emerging molecular information provides critical insights as well as new opportunities to develop molecular and targeted approaches to cancer prevention, detection, and treatment. Although a variety of mutations are often shared by entirely different cancers, some mutational events are unique to specific tumor types. Thus, the molecular blueprints of cancer cells are not only distinct from their normal counterparts, but are also different from each other. Based upon the information obtained from these unique molecular signatures, specific therapeutic strategies can be designed for each tumor type. Instead of using the conventional cytotoxic agents, which are indiscriminately toxic to cancer as well as normal cells, abnormalities identified in specific signal transduction pathways in specific tumors might be targeted. Examples would include use of reagents that interfere with ligand/receptor interactions in those tumors where autocrine or paracrine pathways involving specific growth factors and their receptors are activated. Sometimes structurally altered receptors are present at the cell surface as unique antigens providing an attractive target for immunotherapy using monoclonal antibodies. These receptors may also be used for the precise delivery of conjugated toxic substances to tumor cells.

Dysregulation of the various components of cell cycle is probably the most prominent feature of cancer cell pathology and is an obvious target of antiproliferation strategies. Blocking the progression of cell cycle will be most effective by an appropriate strategic design targeting the right activators or inhibitors of CDKs. Progression of cell cycle is regulated at multiple levels and by multiple proteins with very diverse and sometimes opposing influences. An excellent example is the p53 protein, which influences cell proliferation, cell death, and DNA-damage control. The status of p53 (mutant vs. non-mutant) therefore determines the cancer cell's responsiveness to therapy.

Besides using cell proliferation as a target to limit tumor growth, anticancer strategies can be directed to induce cell suicide or to limit the life span of cells. Much effort is being directed to understand multiple caspase activation pathways and the mechanisms that regulate caspases. Promising approaches will include blocking the expression and/or function of inhibitors of apoptosis such as members of Bcl-2 family, IAPs, and survivin or upregulating activators of caspases such as Apaf and Bcl-x. The enzyme telomerase is upregulated in 90% of human malignancies and, more importantly, in 30% of premalignant lesions. Telomerase can therefore serve a dual purpose; it can be used as a biomarker for diagnostic and possibly prognostic information and it can also serve as a therapeutic target, especially in those cancers where it is detected in preneoplastic stages.

One of the popular therapeutic strategies targets the vasculature of tumors. In solid tumors, formation of new blood vessels occurs by a process of sprouting from preexisting vessels and is known as angiogenesis. During the process of angiogenesis, endothelial cells, which are the principle cell type of blood vessels, proliferate and migrate to form

new vasculature. Thus, a tumor's vasculature is composed of normal endothelial cells, which, in contrast to tumor cells, would be less prone to become drug resistant when targeted by antiangiogenic therapies. A whole battery of growth factors/growth factor receptors, adhesion receptors, and proteases/protease inhibitors participate in the process and lend themselves as potential targets for antiangiogenic strategies. Angiogenesis is very likely the end result of the interactions between multiple pro- and antiangiogenic factors. Some tumors might turn on angiogenesis by upregulating specific proangiogenic growth factors, while in others, it may be the consequence of loss of angiogenesis inhibitors. Accordingly, the information on the angiogenic profile of a particular tumor is crucial for designing most direct and effective angiogenic therapies.

Another concept that has gained tremendous momentum in cancer genetics is the functional significance of genetic polymorphisms in the development of diseases including cancer. Single nucleotide polymorphisms (SNPs) are normal variations in the DNA that make each individual unique. SNPs occur in approximately one in 1000 bases along the human genome. These variations from the consensus provide information as to why one person is susceptible to a disease while another resists it. Thus, it appears that an individual's genotype has an enormous bearing on his or her susceptibility to cancer. This concept now extends far beyond the germline mutations of tumor suppressor genes in a relatively small percentage of the population that predisposes it to cancer. One of the most recent approaches in cancer genetics is the idea of the use of SNPs in tracing genetic variations that contribute to increased susceptibility to cancer. Other uses of SNPs will be in determining the penetrance of germline mutations (in other words, whether individuals with heritable mutations will indeed get cancer), in getting information on what gene environment interactions facilitate the incidence of cancer, and also in tracing the genetic variations that contribute not only to diseases but also to sensitivities to various drugs.

The remarkable progress in the molecular dissection of carcinogenesis is only paralleled by the development of highly sophisticated technologies to analyze and diagnose the molecular alterations with unprecedented sensitivity and specificity. The recent advances, both in the molecular insights into various cancers and emergence of sophisticated high-throughput technologies, enable us to embark on new and formidable initiatives such as fingerprinting pure populations of cancer cells through various stages of their development. For example, Laser Capture Microdissection (LCM) is a laser-guided technique that allows the isolation of a pure population of cancer cells under direct visualization while retaining the morphology and preserving DNA, RNA, and protein. Thus, when necessary, the technique allows molecular analyses of homogeneous cancer cells, which otherwise make up for a small percentage in a heterogeneous tumor comprised of many other normal cell types.

Another technology that has revolutionized our ability to diagnose, analyze and monitor cancer cells is the gene-chip technology that has the capability of scanning hundreds of thousands of nucleotides simultaneously for point mutations. Arrays of thousands of oligonucleotides of predetermined sequences are deposited on small chips by photolithography, a process of light-directed synthesis, and hybridized to fluorescently labeled probes. The signal is then detected by scanning confocal microscopy. The technique thus makes possible mutation detection in very large genes like BRCA1 in a very short period of time. This technology also has the phenomenal capability of monitoring the expression of thousands of genes in parallel. By using doubly labelled

probes from normal and tumor cells, differential expression between the two cell types can be monitored for genes from multiple signal transduction pathways enabling us to analyze networks and patterns of expression. Furthermore, expression chips are being used as a throughput screening procedure making it possible to analyze literally thousands of tumors simultaneously. Another form of chips, SNPs chips will have use in pharmacogenomics for tailoring drugs to the specific molecular profiles of individual patients.

Expression microarray technology is being used for the molecular profiling of neoplastic lesions and will provide fundamental insights into the molecular alterations that subvert the normal cellular processes. It will uncover specific molecular markers for cancer detection/prevention and will identify molecular differences among tumors, which otherwise appear to be similar or even identical by conventional analyses. It will allow us to determine, at various stages of cancer development, which molecular targets will be useful for a specific cancer and therefore which therapies will be appropriate. Thus, it will provide a basis for a) identifying markers for early detection, b) developing new and more relevant experimental animal model systems in which to model the human disease, and also c) greatly enhance the drug development process.

Despite the remarkable progress in understanding the mysteries of cancer, untangling the genetic events leading to cancer represents an endeavor, which is far from over. However, based on the rapidly emerging molecular information of the biology of the cell and innovative sophisticated technologies, a cautious optimism that someday cancer therapies will be customized to take into account both the molecular diagnosis of cancer and the genetic make-up of the patient seems warranted.

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